Gerard J. O'Reilly

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Appointments

Current

- Associate Professor of Structural Engineering @ Centre for Training and Research on Reduction of Seismic Risk (ROSE Centre), IUSS Pavia, Italy — Dec 2023 – present
- Director, Centre for Training and Research on Reduction of Seismic Risk (ROSE Centre) @ IUSS Pavia, Italy — Mar 2025 – present
- Co-Founder and Partner @ Djura — Risk - Data - Engineering Srl, Pavia, Italy — Jul 2025 – present
- Member of the Doctoral Advisory Board (*Collegio dei Docenti*) @ IUSS Pavia, Italy — May 2024 – present
- Member of the Advisory Board (*Comitato di Coordinamento*) @ EUCENTRE Foundation, Italy — Jan 2024 – present
- Affiliated Researcher @ EUCENTRE Foundation, Italy — Jan 2014 – present

Past

- Assistant Professor of Structural Engineering @ Centre for Training and Research on Reduction of Seismic Risk (ROSE Centre), IUSS Pavia, Italy — May 2019 – Nov 2023
- Visiting Researcher @ Kobori Research Complex, Tokyo, Japan — Nov 2019 – Dec 2019
- Post-doctoral Researcher @ Centre for Training and Research on Reduction of Seismic Risk (ROSE Centre), IUSS Pavia, Italy — Mar 2017 – Apr 2019
- Visiting Researcher @ University of California, Berkeley, USA — Sept 2011 – Aug 2012

Education

- Ph.D in Earthquake Engineering & Engineering Seismology, IUSS Pavia, Italy — 2016
- M.Sc. in Civil Engineering, University of Galway, Ireland — 2013
- B.Eng. in Civil Engineering, University of Galway, Ireland — 2010

Honours & Awards

- Awarded €1,321,705 Starting Grant by the Fondo Italiano per la Scienza (FIS) on "Sensor-driven statistical tools to evaluate risks and manage safety in the built environment"
- 2022 recipient of the Shah Family Innovation Prize by the Earthquake Engineering Research Institute to honour an individual under the age of 35 for creativity, innovation and an entrepreneurial spirit in earthquake risk mitigation and management [Link]
- Awarded 2024 Engineering Structures Best Paper of the Year (Artificial Intelligence in Structural Engineering) for paper on "Next-generation non-linear and collapse prediction models for short to long period systems via machine learning methods"
- Awarded Editor's Featured Paper Award by Engineering Structures for paper on "Next-generation non-linear and collapse prediction models for short to long period systems via machine learning methods"
- Won the 2020 Outstanding Paper Award by the International Association for Bridge and Structural Engineering for the research article: Once upon a Time in Italy: The Tale of the Morandi Bridge [Link]

- Top Reviewer, Bulletin of Earthquake Engineering (2020)
- Awarded "Best Structures Paper" for paper presented at the BCRI2012 conference in Dublin, Ireland
- Awarded "University Scholar Prize" in 2007 and 2008 by the University of Galway in recognition of academic excellence during undergraduate studies
- Part of the research group whose paper was awarded the R.G. Drysdale Award at the 15th Canadian Masonry Symposium for "Out-of-Plane Shake-Table Tests on Unreinforced Masonry Gables"
- Part of the research group whose paper was awarded Honourable Mention for the M.A. Hatzinikolas Award at the 15th Canadian Masonry Symposium for "In-Plane Cyclic Shear-Compression Tests on Stone Masonry Piers Strengthened with Composite- and Fiber-Reinforced Mortars"

Research Projects

Research Project Coordination

- **SONATA: Sensor-driven statistical tools to evaluate risks and manage safety in the built environment** Oct 2025 – Sept 2028

  Webpage: gerardjoreilly.github.io/projects/sonata

  Role: Principal Investigator

  Budget: €1,321,705.00

  Funding: Starting Grant – Fondo Italiano per la Scienza (FIS), Italian Ministry of University and Research (MUR)

  Description: SONATA aims to improve earthquake safety in Italy's built environment, with a particular focus on non-structural elements that significantly impact economic losses and post-earthquake recovery. Using innovative experimental tools including the 9DLAB shaking table, sensors, artificial intelligence, and image recognition, the project develops advanced methods to assess seismic damage and derive accurate fragility functions. These tools are integrated into a prototype system and validated through real-world case studies to support effective risk management and decision-making.

- **Progetto DPC/ReLUIS 2024-2026: WP5.4: Rapid, low impact and integrated retrofitting schemes for bridges** Apr 2024 – present

  Webpage: http://www.reluis.it/

  Role: Coordinator

  Budget: €75,000

  Funding: Italian Civil Protection Department

  Description: The investigation of damage mechanisms in existing bridge structures commonly found throughout Italy and comparison with past damage observations. The objective is to investigate the level of inter- and intra-structure correlation between these typologies to facilitate the more refined regional assessment and prioritisation of required interventions.

- **EFEHR - European Facilities for Earthquake Hazard and Risk** Jul 2024 – present

  Webpage: http://www.efehr.org/start/

  Role: Local Coordinator

  Budget: €8,600

  Funding: European Plate Observing System - European Research Infrastructure Consortium (EPOS-ERIC)

  Description: This project facilitates the EUCENTRE's involvement in advancing earthquake hazard and risk assessment in the European-Mediterranean area as part of the EFEHR consortium's overall efforts.

- **Built Environment Data Thematic Core Service** Oct 2022 – present

  Webpage: https://builtenvdata.eu/

  Role: Coordinator

Funding: Italian Ministry of University and Research (MUR)

Description: The development of a new candidate thematic core service as part of EPOS (European Plate Observing System) related to the built environment. This comprises the development of databases and computational tools that facilitate the collaboration and sharing of data and knowledge according to the FAIR (Findable, Accessible, Interoperable, and Reusable) principles related to buildings and urban infrastructure.

- **Progetto Dipartimento di Eccellenza 2023-2027** Jan 2023 – Dec 2027

Webpage: www.iusspavia.it/it/ricerca/dipartimento-di-eccellenza-2023-2027

Role: Coordinator (Line R1)

Budget: €6,650,000

Funding: Italian Ministry of University and Research (MUR)

Description: The development of a risk-based approach for the sustainable design of structures and infrastructures with particular regard to those of protection from natural disasters and the sustainable development of the nation.

- **ERIES: Engineering Research Infrastructures for European Synergies** Jun 2022 – May 2026

Webpage: www.eries.eu

Role: Project Coordinator

Budget: €11,616,118.24

Funding: European Union Horizon Europe

Description: ERIES provides transnational access to advanced experimental facilities in the fields of structural, seismic, wind and geotechnical engineering, with new and unique infrastructures available for the first time. It allows users to advance frontier knowledge and conduct curiosity-driven research towards the reduction of losses and disruption due to these hazards, the management of their associated risk, and the development of innovative solutions to address them that will contribute to a greener and more sustainable society. The project comprises 13 partners from 8 different countries across Europe and North America, including: IUSS Pavia (Italy, Coordinator), Eucentre Foundation (Italy); University of Patras (Greece); Aristotle University of Thessaloniki (Greece); Laboratório Nacional de Engenharia Civil (LNEC), Lisbon (Portugal); Centre de recherche (CEA) Paris-Saclay (France); University of Bristol (United Kingdom); Institute of Earthquake Engineering and Engineering Seismology (IZIIS) Skopje (N. Macedonia); University of Genova (Italy); Western University (Canada); Eindhoven University of Technology (TU/e), Eindhoven (Netherlands); Centre Scientifique et Technique du Bâtiment (CSTB) Nantes (France); European Laboratory for Structural Assessment at the Joint Research Centre (JRC), Ispra (Italy).

- **ROSSINI: Progettazione, realizzazione e sperimentazione di un sistema prototipale di navigazione risk-aware per la gestione e mitigazione del RischiO SiSmico in industrie a rischio di INcidente rIlevante** Jun 2020 – Nov 2022

Webpage: www.progetto-rossini.it

Role: Principal Investigator and Coordinator

Budget: €333,300

Funding: Italian National Institute for Insurance against Accidents at Work (INAIL)

Description: ROSSINI placed itself within the overall field of seismic risk-aware navigation systems. It used of a sensor array in tandem with and customisable library of fragility functions and numerical models for the integrated risk-aware navigation in industrial plants at risk of NaTech accidents, using an industrial plant facility in Italy as its pilot study. The integration of structural and environmental risks estimated and measured in different ways via a sensor array network was presented as part of an integrated risk identification and evaluation (RIE) online module. This module was used to combine and map the possible risks spatially within an industrial plant's layout and shown how this information can then be used to not only compute the safest path to safety for a worker located within such a plant but also how mobile communications can be used to aid and guide them in different scenarios.

- **Progetto Dipartimento di Eccellenza 2018-2022** Jan 2019 – Jun 2022

Webpage: www.iusspavia.it/it/ricerca/dipartimento-di-eccellenza-2018-2022

Role: Deputy Coordinator (Line 3)

Budget: €8,205,100

Funding: Italian Ministry of University and Research (MUR)

Description: The research activities associated with Line 3 envisaged a complete review of available seismic design and assessment methods currently employed. It focussed on issues related to problems with current code-based approaches and aimed to incorporate avant-garde approaches such as risk-targeted and loss-driven approaches in order to promote and more optimal use of available resources in the future with the overall goal of reducing seismic risk.

- **INFRA-NAT – Increased Resilience of Critical Infrastructure under Natural and Human-induced Hazard** Jan 2018 – Dec 2019

Webpage: www.infra-nat.eu

Role: Deputy Coordinator and Research Collaborator

Budget: €762,839.05

Funding: European Union Directorate-General for European Civil Protection and Humanitarian Aid Operations (DG-ECHO)

Description: The objective of the INFRA-NAT project was to assess the seismic vulnerability of roadway networks taking into account ageing and deterioration effects of their bridge structures. It focussed on the collection of structure data in Italy, Israel and North Macedonia with the aim of engaging local practitioners and authorities to encourage the adoption of such assessment and prioritisation methodologies.

Research Project Collaboration

- **ERIES-PASFIT: Long-Term Performance Assessment of Base Isolated Buildings through Field Testing** Jan 2024 – Dec 2024

Role: Research Collaborator & Task Coordinator

Funding: ERIES Transnational Access Joint Research Funding Mechanism

Description: ERIES- PASFIT will conduct full-scale in-situ testing of buildings, base isolated via laminated rubber bearings and friction pendulum (FP) systems that have been in service for approximately 15 years. The results of these field tests will be the first of their kind, and will be used to fill important research gaps pertaining to the characterisation of the response and performance of base isolated buildings, including aspects of system-level response in conjunction with ageing, deterioration and, more generally, variability of the mechanical properties of the isolation devices. Thus, this will have important implications on base isolation design, assessment and modelling strategies.

- **ERIES-RACKSLIDE: Content sliding investigations for pallet racking systems** Jan 2024 – Dec 2024

Role: Research Collaborator & Task Coordinator

Funding: ERIES Transnational Access Joint Research Funding Mechanism

Description: Racking systems form the backbone of modern logistics and goods supply chains, lying at the core of every physical product route from manufacturer to consumer. Rack-supported and rack-containing warehouses come in various shapes and sizes, universally constructed from cold-formed steel frames that range from 4m to more than 25m of height. The vast majority stores palletized goods that stay put only by the force of gravity and friction. This is a potentially vulnerable situation that has been severely tested by recent earthquakes, resulting in disruptions of operation and some spectacular collapses. Presently, there are evident gaps in the slide-proof design of racking systems and even in their sliding assessment, both at the professional and the academic level. ERIES-RACKSLIDE brings together 7 universities, one association of manufacturers, and 5 supporting industrial partners with active interest in researching pallet sliding, and proposing the usage of the 9DLAB facility to conduct innovative investigations that will form the state-of-art in content- structure-sliding interaction and inform the future of EN16681, the seismic rack-design standard.

- **ERIES-SUPREME: Seismic oUt-of-Plane REsponse of Masonry gablEs** Jan 2024 – Dec 2024

Role: Research Collaborator & Task Coordinator

Funding: ERIES Transnational Access Joint Research Funding Mechanism

Description: Low-rise masonry buildings in Europe often have unreinforced masonry walls and masonry gables supporting pitched roofs. These buildings are common in seismic-prone areas and are particularly vulnerable to earthquakes. The masonry gables are especially susceptible due to their poor connections to the roof structure, positioning at the top of the building, and interaction with flexible roof diaphragms. However, there is limited experimental data on the seismic performance of these gables. This project aims to improve understanding by conducting experiments with easily replicable conditions for numerical simulations. The data will be used to refine existing models and develop new tools for assessing gable performance. A blind-prediction contest will be held to encourage calibration, and the results will be published in a peer-reviewed journal to improve assessment guidelines.

- **ERIES-ENFRAG: ENhancing state-dependent FRAGility through experimentally validated Energy-Based Approaches** May 2023 – Jul 2024

Role: Research Collaborator & Task Coordinator

Funding: ERIES Transnational Access Joint Research Funding Mechanism

Description: ERIES-ENFRAG advances state-dependent earthquake fragility assessment methodologies. The project focuses on masonry infill walls experiencing cumulative states of damage due to combinations of in- plane (IP) and out-of-plane (OOP) actions, commonly quantified through two different peak-based engineering demand parameters (EDPs). ERIES-ENFRAG explores the experimental validation of hysteretic energy-based fragility assessment approaches, which are: 1) currently based only on analytical and/or numerical validations; 2) only considering one type of action/damage mechanism. ERIES-ENFRAG will pave the way for similar cumulative-damage tests for different structures/structural components while providing experimental data on the IP and OOP response of masonry infills. ERIES-ENFRAG aims at a robust consideration of damage-accumulation, particularly relevant in mainshock-aftershock conditions, and will also shed further light on the appropriateness of IP and OOP loading protocols used in experimental testing.

- **ERIES-RESTORING: REtrofitting of STOne masonRy using INnovative Grid- based composites** May 2023 – Jul 2024

Role: Research Collaborator & Task Coordinator

Funding: ERIES Transnational Access Joint Research Funding Mechanism

Description: The seismic retrofitting of existing masonry buildings has become a priority in seismically prone countries with significant cultural heritage. The proposed project aims to research the effectiveness of Composite Reinforced Mortars (CRM) in strengthening rubble stone masonry buildings. Full-scale tests will be conducted on piers with CRM applied to assess their behavior compared to non-retrofitted piers. The project will also investigate different aspect ratios and load rates. The data collected will contribute to future design guidelines for CRM as a strengthening solution, reducing vulnerability and losses after earthquakes.

- **Progetto DPC/ReLUIS 2019-2021: WP5.1: Assessment of integrated seismic retrofitting schemes for buildings** Jan 2020 – Jan 2022

Role: Research Collaborator

Budget: €75,000

Funding: Italian Civil Protection Department

- **Progetto DPC/ReLUIS 2019-2021: WP5.4: Rapid, low impact and integrated retrofitting schemes for bridges** Jan 2019 - Jan 2022

Role: Task Coordinator and Collaborator

Budget: €75,000

Funding: Italian Civil Protection Department

- **Investigation of Seismic Deformation Demand, Capacity and Control in a Novel Self-Centring Steel Braced Frame** Jan 2019 – Jun 2021

Role: Research Collaborator

Description: SERA: Seismology and Earthquake Engineering Research Infrastructure Alliance for Europe (EU Horizon 2020)

- **Joint Research: Comparison in Performance of Base-Isolated Structures between Japan and Italy @ Kobori Research Complex** Nov 2019 – Dec 2019

  Role: Lead Researcher
  Budget: €6,850 (822,000 JPY)
  Funding: Kobori Research Complex, Tokyo, Japan

- **ReLUIS/DPC Line 7: Displacement-based seismic loss assessment** May 2016 – Dec 2018

  Role: Lead Researcher
  Budget: €475,250.00
  Funding: Italian Civil Protection Department

- **Progetto Scuole** May 2015 – Dec 2016

  Role: Lead Researcher
  Budget: €1,500,000
  Funding: Centro Geomorfologia Integrata Per L'Area Del Mediterraneo

- **DiSTEEL: Displacement-Based Seismic Design of Steel Moment Resisting Frame Structures** Feb 2014 – May 2014

  Role: Researcher Collaborator
  Funding: European Commission Research Fund of Coal and Steel

- **BRACED: Improved European design and assessment methods for concentrically-braced frames** Sep 2012 - Aug 2013

  Role: Researcher Collaborator
  Funding: SERIES: Seismic Engineering Research Infrastructures for European Synergies (European Commission FP7)

People

Research Associates

1. **Daniele Sivori**

   Period: Nov 2025 - present

2. **Volkan Özsaraç**

   Period: Oct 2025 - present (Postdoctoral Researcher: May 2024 - Sept 2025)

3. **Davit Shahnazaryan**

   Period: Oct 2025 - present (Postdoctoral Researcher: Dec 2021 - Sept 2025)

PhD Students

1. **Marco Todaro**

   Topic: Sensor-driven statistical tools to evaluate risks and manage safety in the built environment
   Period: Oct 2025 - Oct 2028 (expected)
   Role: Principal advisor (Co-advisors: Daniele Sivori, Davit Shahnazaryan)

2. **Mauricio Guamán Naranjo**

   Topic: Sensor-driven statistical tools to evaluate risks and manage safety in the built environment
   Period: Oct 2025 - Oct 2028 (expected)
   Role: Principal advisor (Co-advisors: Daniele Sivori, Volkan Ozsarac)

3. **Vitor A. Monteiro**

   Topic: Regional seismic risk assessment incorporating spatial correlation of next-generation intensity measures

Period: Oct 2024 - Oct 2027 (expected)

Role: Principal advisor

4. **Tomas Mejia Saldarriaga**

   Topic: Regional seismic risk assessment of structures incorporating building-to-building damage correlations

   Period: Oct 2024 - Oct 2027 (expected)

   Role: Principal advisor

5. **Serkan Hasanoğlu**

   Topic: Seismic risk analysis of Turkish residential building taxonomies through simulated design

   Period: Oct 2022 - Oct 2026 (expected)

   Role: Principal advisor

6. **José Poveda**

   Topic: Investigating impacts of ground motion and hazard definitions in the seismic analysis and capacity design of reinforced concrete wall buildings

   Period: Oct 2022 - Oct 2026 (expected)

   Role: Principal advisor

7. **Savvinos Aristeidou**

   Title: Advances in the practical application of next-generation intensity measures for efficient seismic risk assessment [PDF]

   Period: Oct 2020 - Dec 2024

   Role: Principal advisor

8. **Al Mouayed Bellah Nafeh**

   Title: Advancements in Risk- and Loss-Based Methodologies for Large-Scale Assessment of Non-Ductile Infilled Reinforced Concrete Buildings [PDF]

   Period: Oct 2019 - Nov 2023

   Role: Principal advisor

9. **Andrés Abarca Jimenez**

   Title: Regional Seismic Risk Assessment for Integrated Loss–Based Prioritisation of Bridge Portfolios [PDF]

   Period: Sept 2017 - Apr 2022

   Role: Co-advisor (Principal advisor: Ricardo Monteiro)

10. **Davit Shahnazaryan**

    Title: Integrated Performance-Based Seismic Design: Traversing Affordances for Practical Implementation [PDF]

    Period: Sept 2017 - Nov 2021

    Role: Principal advisor (Co-advisor: Ricardo Monteiro)

MSc Students

1. **Maria Agni Ladeni**

   Title: Comparative analysis of exposure databases and portfolio diversification in catastrophe modeling for (Re)insurance applications in Italy

   Period: Feb 2025 – Sept 2025

   Role: Co-advisor (Principal advisor: Dr. Matt J. Fox)

2. **Wendwesen Fekede**

Title: Post lateral-torsional buckling behaviour of a simply-supported beam subjected to midspan point load [PDF]

Period: Feb 2025 – Sept 2025

Role: Co-advisor (Principal advisor: Prof. Richard Wiebe)

3. **Tomas Mejia Saldarriaga**

   Title: Quantifying the impacts of structure-to-structure damage correlation on regional seismic risk assessment

   Period: Feb 2024 – Sept 2024

   Role: Principal advisor

4. **Letizia Palamara**

   Title: Italian guidelines for the risk classification and management of existing bridges: a case study implementation and evaluation [PDF]

   Period: Feb 2021 - Oct 2021

   Role: Co-advisor (Principal advisor: Gian Michele Calvi)

5. **Federico Damiani**

   Title: Seismic design and loss assessment of base-isolated structures using a PBEE framework [PDF]

   Period: Feb 2020 - Dec 2020

   Role: Co-advisor (Principal advisor: Ricardo Monteiro)

6. **Savvinos Aristeidou**

   Title: A Performance-Based Assessment of a Case Study Bridge and Impact of Retrofitting Interventions [PDF]

   Period: Feb 2020 – Sept 2020

   Role: Co-advisor (Principal advisor: Gian Michele Calvi)

7. **Carlos Andres Mora Castrillo**

   Title: Risk-Consistency of Force-Based and Displacement-Based Design of Reinforced Concrete Moment Frames [PDF]

   Period: Sept 2018 – Feb 2019

   Role: Co-advisor (Principal advisor: Ricardo Monteiro)

8. **Onur Deniz Akan**

   Title: Displaced Shapes of Infilled RC Frames for Displacement-Based Design and Assessment [PDF]

   Period: Sept 2018 – Mar 2019

   Role: Co-advisor (Principal advisor: Ricardo Monteiro)

9. **Jose Rafael Leone Villalba**

   Title: Evaluation of Risk-Consistency of Force-Based and Displacement-Based Design of RC Wall Buildings [PDF]

   Period: Sept 2018 – Feb 2019

   Role: Co-advisor (Principal advisor: Ricardo Monteiro)

10. **Wilson Carofilis**

    Title: Seismic Assessment and Retrofit of an Existing Reinforced Concrete School Building in Italy [PDF]

    Period: Sept 2017 – Feb 2018

    Role: Principal advisor (Co-advisors: Andre Filiatrault, Daniele Perrone, Ricardo Monteiro)

Synergistic Activities

Keynote and Invited Lectures

- Invited theme lecture on "Advancements in seismic risk assessment of existing structures: insights from analytical models and empirical earthquake data" at the 10th Turkish Conference on Earthquake Engineering in Istanbul, Türkiye — September 2025
- Invited lecture on "Embracing Digital Tools and Open Science for Engineering Innovation" at the 14th International Conference on Structural Analysis of Historical Constructions (SAHC 2025), Lausanne, Switzerland — September 2025
- Invited seminar on "The role of non-structural element vulnerability and sensor-driven risk assessment in post-earthquake recovery and decision-making: lessons from Japanese field data" at the University of Bristol, UK — May 2025
- Invited lecture on "A consistent framework for seismic risk assessment, classification and decision making for existing buildings" at the 1st NORISK Workshop "The role of structural safety in the design and assessment of constructions and infrastructures", Venice, Italy — June 2025
- Invited theme lecture on "The Built Environment Data Framework for Simulated Design and Vulnerability Modelling in Earthquake Engineering" at the 9th International Course on Seismic Analysis of Structures Using OpenSees in Lausanne, Switzerland — April 2025
- Invited lecture on "Built environment data for multi-hazard vulnerability models within EPOS" at the EFEHR Scientific Session 2024 organised as part of the 102nd Journées Luxembourgeoises de Géodynamique (JLG) in Luxembourg — November 2024
- Invited lecture on "Quantifying fragility functions for non-ductile infilled reinforced concrete buildings from past earthquakes: analytical models versus empirical data" at the 3rd Structural Engineering Workshop by the University of Palermo, Italy — July 2024
- Keynote lecture on "European Research Synergies Towards Loss and Risk-Driven Mitigation Approaches" at the 20th International Symposium of the Macedonian Association of Structural Engineers from 28-29 September 2023 in Skopje, North Macedonia — September 2023
- Invited lecture on "Myths and fallacies in performance-based earthquake engineering: Ode to Nigel" at the 49th Risk, Hazard and Uncertainty Workshop in Hydra, Greece — June 2023
- Invited lecture on "Valutazione e gestione del rischio NaTech negli stabilimenti PIR" at the symposium organised by INAIL in Rome, Italy on the topic of "La ricerca nel campo della gestione del rischio NaTech da sisma" — May 2023
- Invited webinar to the 2nd RICH Europe Webinar on Transnational and Virtual Access Opportunities on the topic of "ERIES – Engineering Research Infrastructures for European Synergies" — April 2023
- Invited webinar to the Italian Association for Wind Engineering Young (ANIV - Associazione Nazionale per Ingegneria del Vento - Giovani) on the topic of "ERIES research infrastructure network: how to apply for transnational access grants" — November 2022
- Invited talk on "Developments in performance-based seismic design and assessment: an Italian perspective" at the University of Canterbury, New Zealand — October 2019

Professional Memberships and Qualifications

- Member of the European Geosciences Union (EGU) — 2023 – present
- Approved by the Italian Ministry of Universities and Research's National Scientific Committee as qualified to hold the rank of Full Professor in Structural and Geotechnical Engineering — 2025 - 2037
- Approved by the Italian Ministry of Universities and Research's National Scientific Committee as qualified to hold the rank of Associate Professor in Structural and Geotechnical Engineering — 2022 - 2031

- Member of the Earthquake Engineering Research Institute (EERI) 2022 – present

Coordination and Editorial Roles

- Editorial Board Member in the role of Associate Editor for Earthquake Spectra journal 2025 - present
- Director of the Centre for Training and Research on Reduction of Seismic Risk (ROSE Centre) at IUSS Pavia, Italy, organising regular online seminars available on YouTube and maintaining a constant social media presence on LinkedIn to disseminate the research centre's activities (Deputy Director 2020-2025) 2025 - present
- Member of the Editorial Board of Geohazards and Georisks for the journal Frontiers in Earth Science 2020 – present
- Member of the Editorial Board of Earthquake Engineering for the journal Frontiers in Built Environment 2020 – present

Conference and Workshop Organisation

- Chair of the Organising Committee for the ERIES International Workshop 2026 (ERIES-IW2026), hosted by EUCENTRE Foundation, Pavia, Italy 2026
- Member of the Scientific Committee for the 18th European Conference on Earthquake Engineering (18ECEE), Berlin, Germany 2026
- Chair of the Organising Committee for the ERIES International Workshop 2025 (ERIES-IW2025), hosted by LNEC, Lisbon, Portugal 2025
- Organising Committee Member, 18th World Conference on Earthquake Engineering (WCEE2024), Milan, Italy 2024

International Conference Participation

- SPONSE 2026 - 6th International Workshop on Seismic Performance of Non-Structural Elements, Tokyo, Japan 2026
- COMPDYN 2025 - 10th ECCOAS Thematic Conference on Computational Methods in Structural Dynamics and Earthquake Engineering, Rhodes, Greece 2025
- 10th Turkish Conference on Earthquake Engineering, Istanbul, Türkiye 2025
  - Invited theme lecturer
- XX Convegno ANIDIS - Associazione Nazionale Italiana di Ingegneria Sismica, Assisi, Italy 2025
  - Special session co-chair and speaker
- SAHC 2025 - 14th International Conference on Structural Analysis of Historical Constructions, Lausanne, Switzerland 2025
  - Invited speaker
- EPOS Seismology & Geo-INQUIRE Workshop, Athens, Greece 2025
- EFEHR Stakeholder Workshop: Connecting with Insurance and Cat-modelling Professionals, University College London, UK 2025
- EGU General Assembly 2025, Vienna, Austria 2025
- 3rd Croatian Conference on Earthquake Engineering (CroCEE), Split, Croatia 2025
- WCEE2024 - 18th World Conference on Earthquake Engineering, Milan, Italy 2024
  - Organiser and chairman for the following technical sessions:
    - Economic loss in seismic areas: from empirical data to prescriptive guidance for mitigation
    - ERIES: Advancing frontier knowledge in earthquake engineering through laboratory testing
    - Characterizing seismic input for engineering applications

- 39th General Assembly of the European Seismological Commission (ESC), Corfu, Greece 2024
- EERI Annual Meeting, Seattle, USA 2024
- SECED 2023 Conference - Society for Earthquake and Civil Engineering Dynamics, Cambridge, UK 2023
- MASE - 20th International Symposium of the Macedonian Association of Structural Engineers, Skopje, North Macedonia 2023
  - Keynote speaker
- ICASP14 - 14th International Conference on Applications of Statistics and Probability in Civil Engineering, Dublin, Ireland 2023
  - Chairman for general session on "Novel challenges in performance-based seismic design and seismic performance assessment of structures"
- 49th Risk, Hazard and Uncertainty Workshop, Hydra, Greece 2023
- Earthquake Engineering Research Institute Annual Meeting, San Francisco, USA 2023
- ANIDIS - Associazione Nazionale Italiana di Ingegneria Sismica 2022, Turin, Italy 2022
- 3ECEES - 3rd European Conference on Earthquake Engineering & Seismology, Bucharest, Romania 2022
- ASME 2022 Pressure Vessels & Piping Conference, Las Vegas, USA 2022
- ICONHIC 2022 - 3rd International Conference on Natural Hazards & Infrastructure, Athens, Greece 2022
- 17WCEE - 17th World Conference on Earthquake Engineering, Sendai, Japan 2021
- COMPDYN 2021 - 8th ECCOAS Thematic Conference on Computational Methods in Structural Dynamics and Earthquake Engineering, Athens, Greece 2021
- QuakeCoRE Annual Meeting, Nelson, New Zealand 2019
- ICASP13 - 13th International Conference on Applications of Statistics and Probability in Civil Engineering, Seoul, South Korea 2019
- COMPDYN 2019 - 7th ECCOAS Thematic Conference on Computational Methods in Structural Dynamics and Earthquake Engineering, Hersonissos, Greece 2019
- 4SPONSE - The Fourth International Workshop on Seismic Performance of Non-Structural Elements, Pavia, Italy 2019
- 16ECEE - 16th European Conference on Earthquake Engineering, Thessaloniki, Greece 2018
  - Chairman for general session on "Reinforced Concrete Structures IX"
- COMPDYN 2017 - 6th International Conference on Computational Methods in Structural Dynamics and Earthquake Engineering, Rhodes, Greece 2017
- IMAC-XXXV Conference & Exposition on Structural Dynamics, Orange County, USA 2017
- 16WCEE - 16th World Conference on Earthquake Engineering, Santiago, Chile 2017
  - Chairman for general session on "Seismic Design and Analysis of Reinforced Concrete Buildings"
- COMPDYN 2015 - 5th ECCOAS Thematic Conference on Computational Methods in Structural Dynamics and Earthquake Engineering, Hersonissos, Greece 2015
- 2ICSA - 2nd International Conference on Structures and Architecture, Guimaraes, Portugal 2013
- BCRI2012 - Bridge and Concrete Research in Ireland, Dublin, Ireland 2012
- 15WCEE - 15th World Conference on Earthquake Engineering, Lisbon, Portugal 2012

Teaching

Academic Year 2025/2026

- **Dynamics of Structures**

  Role: Lecturer (22 hours)
  Period: September – October 2025
  Programme: Master Degree in Civil Engineering for Mitigation of Risk from Natural Hazards
  Institute: Università di Pavia and IUSS Pavia, Italy

- **Performance-Based Earthquake Engineering**

  Role: Lecturer (48 hours)
  Period: January 2026
  Programme: Doctoral Programme Course in Earthquake Engineering
  Institute: IUSS Pavia, Italy

- **Dynamic Analysis of Structures**

  Role: Lecturer (31 hours)
  Period: April 2026
  Programme: Master Degree in Earthquake Engineering
  Institute: Università di Pavia and IUSS Pavia, Italy

- **Understanding Catastrophe Risk: Modelling, Management, and Reinsurance**

  Role: Lecturer (10 hours)
  Period: April 2026
  Programme: Advanced undergraduate course (*Corsi Ordinari*)
  Institute: IUSS Pavia, Italy

Academic Year 2024/2025

- **Dynamics of Structures**

  Role: Lecturer (22 hours)
  Programme: Master Degree in Civil Engineering for Mitigation of Risk from Natural Hazards
  Institute: Università di Pavia and IUSS Pavia, Italy

- **Performance-Based Earthquake Engineering**

  Role: Lecturer (48 hours)
  Programme: Doctoral Course in Earthquake Engineering
  Institute: IUSS Pavia, Italy

- **Risk, Reliability and Catastrophe Modelling**

  Role: Lecturer (10 hours)
  Programme: Advanced undergraduate course (*Corsi Ordinari*)
  Institute: IUSS Pavia, Italy

Academic Year 2023/2024

- **Dynamics of Structures**

  Role: Lecturer (22 hours)
  Programme: Master Degree in Civil Engineering for Mitigation of Risk from Natural Hazards
  Institute: Università di Pavia and IUSS Pavia, Italy

- **Performance-Based Earthquake Engineering**

  Role: Lecturer (37 hours)
  Programme: Doctoral Course in Earthquake Engineering
  Institute: IUSS Pavia, Italy

Academic Year 2022/2023

- **Performance-Based Earthquake Engineering**

  Role: Lecturer (37 hours)
  Programme: Doctoral Course in Understanding and Managing Extremes
  Institute: IUSS Pavia, Italy

Academic Year 2020/2021

- **Laboratorio di Tecnica delle Costruzioni (*in Italian*)**

  Role: Lecturer (26 hours)
  Programme: Laurea Magistrale in Ingegneria Edile-Architettura
  Institute: Università di Pavia, Italy

Academic Year 2019/2020

- **Fundamentals of Seismic Design**

  Role: Guest Lecturer (4 lectures, 12 hours)
  Programme: Master Degree in Civil Engineering for Mitigation of Risk from Natural Hazards
  Institute: Università di Pavia and IUSS Pavia, Italy

Academic Year 2018/2019

- **Fundamentals of Seismic Design**

  Role: Guest Lecturer (4 lectures, 12 hours)
  Programme: Master Degree in Civil Engineering for Mitigation of Risk from Natural Hazards
  Institute: Università di Pavia and IUSS Pavia, Italy

- **Performance-Based Earthquake Engineering**

  Role: Guest Lecturer (1 lecture, 3 hours)
  Programme: Master Program of Structural Engineering
  Institute: Universidad de las Fuerzas Armadas – ESPE, Ecuador

Academic Year 2017/2018

- **Fundamentals of Seismic Design**

  Role: Guest Lecturer (4 lectures, 12 hours)
  Programme: Master Degree in Civil Engineering for Mitigation of Risk from Natural Hazards
  Institute: Università di Pavia and IUSS Pavia, Italy

Academic Year 2016/2017

- **Seismic Design of Building Structures**

  Role: Teaching Assistant
  Lecturer: Ricardo Monteiro
  Programme: MSc Programme in Civil Engineering
  Institute: University of Stellenbosch, South Africa

Academic Year 2013/2014

- **Seismic Design of Steel Structures**

  Role: Teaching Assistant
  Lecturer: Roberto Leon
  Programme: ROSE/MEEES MSc Programme
  Institute: Università di Pavia and IUSS Pavia, Italy

Publications

Journal Papers

[1] V. A. Monteiro and G. J. O'Reilly, "Notes on spatial correlation for average spectral acceleration: Direct and indirect approaches," *Earthquake Spectra*, 2026. DOI: 10.1002/esp4.70049.

[2] V. A. Monteiro and G. J. O'Reilly, "A review of ground motion correlation modelling for regional seismic risk analysis," *Bulletin of Earthquake Engineering*, Feb. 2026, ISSN: 1570-761X. DOI: 10.1007/s10518-026-02377-0.

[3] S. Sharma, N. Damiani, M. Bertassi, *et al.*, "Experimental data from out-of-plane shake-table tests on unreinforced masonry gables," *Earthquake Spectra*, Oct. 2025, ISSN: 8755-2930. DOI: 10.1177/87552930251378248.

[4] S. Hasanoğlu, V. Ozsarac, and G. J. O'Reilly, "A model for the simulated design of turkish rc frame buildings in seismic vulnerability analysis," *Bulletin of Earthquake Engineering*, vol. 23, pp. 6829–6856, 15 Dec. 2025, ISSN: 1570-761X. DOI: 10.1007/s10518-025-02301-y. [Online]. Available: https://link.springer.com/10.1007/s10518-025-02301-y.

[5] T. Mejia and G. J. O'Reilly, "Quantifying the impacts of incorporating damage correlation on scenario-based regional seismic assessment," *Earthquake Engineering & Structural Dynamics*, vol. 55, pp. 94–111, 1 Jan. 2026, ISSN: 0098-8847. DOI: 10.1002/eqe.70069. [Online]. Available: https://onlinelibrary.wiley.com/doi/10.1002/eqe.70069.

[6] P. Calvi, A. Rapone, T. Becker, *et al.*, "Dynamic field testing of a 15-year-old friction pendulum base-isolated residential building," *Soil Dynamics and Earthquake Engineering*, vol. 200, p. 109 802, Jan. 2026, ISSN: 02677261. DOI: 10.1016/j.soildyn.2025.109802.

[7] J. Poveda and G. J. O'Reilly, "Impact of ground motion record selection on higher-mode responses in rc wall buildings: Comparing uniform hazard and conditional spectra approaches," *Bulletin of Earthquake Engineering*, vol. 23, pp. 5379–5398, 13 Oct. 2025, ISSN: 1570-761X. DOI: 10.1007/s10518-025-02259-x. [Online]. Available: https://link.springer.com/10.1007/s10518-025-02259-x.

[8] S. Baddipalli, R. Gentile, G. J. O'Reilly, D. Shahnazaryan, and M. Z. Esteghamati, "Impact of modeling decisions on seismic loss and fragility assessment of steel buildings," *Earthquake Spectra*, 2025, ISSN: 19448201. DOI: 10.1177/87552930251360751.

[9] V. Ozsarac, N. Pereira, H. Mohamed, X. Romão, and G. J. O'Reilly, "The built environment data framework for simulated design and vulnerability modelling in earthquake engineering," *Earthquake Engineering & Structural Dynamics*, May 2025, ISSN: 0098-8847. DOI: 10.1002/eqe.4378. [Online]. Available: https://onlinelibrary.wiley.com/doi/10.1002/eqe.4378.

[10] Y. Fukutomi, G. O'Reilly, M. Nakashima, K. Kanda, A. Kusaka, and S. Ogasawara, "Three benefits of "caution (yellow tag)" in shm-driven condition assessment of buildings: Eight years experience with market-based shm," *Earthquake Spectra*, vol. 41, pp. 979–994, 1 Feb. 2025, ISSN: 8755-2930. DOI: 10.1177/87552930241296377. [Online]. Available: https://journals.sagepub.com/doi/10.1177/87552930241296377.

[11] J. Poveda and G. J. O'Reilly, "Seismic loss assessment of existing hotel building in ecuador," *Earthquake Spectra*, vol. 41, pp. 955–978, 1 Feb. 2025, ISSN: 8755-2930. DOI: 10.1177/87552930241299356. [Online]. Available: https://onlinelibrary.wiley.com/doi/10.1177/87552930241299356.

[12] S. Aristeidou, D. Shahnazaryan, and G. J. O'Reilly, "Correlation models for next-generation amplitude and cumulative intensity measures using artificial neural networks," *Earthquake Spectra*, vol. 41, pp. 851–875, 1 Feb. 2025, ISSN: 8755-2930. DOI: 10.1177/87552930241270563. [Online]. Available: https://journals.sagepub.com/doi/10.1177/87552930241270563.

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